75/117 = 64%

CSCI 3350 Software Engineering II **Test 1 K** **Name** Brandon Buchanan

Part 1 – Factoid Questions **(2 points each)**

24/34

Read carefully, these questions are not designed to be deliberately misleading, but words have meaning.

1. The three goals of a successful project are acceptable quality, within budget, on schedule.
2. Anything that adversely affects the three goals of a successful project is a project failure -2. risk
3. **True** or **False?** The purpose or goal of any oral presentation is effective communications.
4. Milestones are major events that occur during the course of the project at a single time, with no associated duration.
5. The starting point for the project schedule is the project kickoff ... WBS
6. COCOMO is an automated cost-model developed by Barry Boehm.
7. **True** or **False?** It is okay to read the script for a technical presentation if the topic is too complicated for you to remember.
8. A good low tech means for implementing anonymous risk reporting is the old-fashioned suggestion box.
9. Two categories of software size metrics are lines of code, and FFP function oriented. -2
10. **True** or **False?** The second level in a Work Breakdown Structure contains the project team leaders.
11. Risk Management consists of Risk Assessment Activities and Risk Control Activities. Three elements of Risk Control are -2 risk mgt. planning, -2 risk resolution, and -2 risk monitoring.
12. **True** or **False?** With regard to oral presentation visual aids, the 8H rule is a useful rule of thumb for determine the number of slides in your presentation.

Part 2 – Short answer (4 points each)

Brief, to the point answers, **maximum of 5 statements**, complete sentences not required.

20/40

1. Your boss shows you a copy of the Cost Breakdown Structure for the project, on which you are currently working. The Work Breakdown Structure is a 37-page Word document. The project total effort is 6 man-months. Your boss asks you to review the Work Breakdown Structure and give her your opinion at the weekly staff meeting tomorrow. What is your initial reaction?

For just 6 man-months of work, there are alot of pages in the Work Breakdown Structure. More than likely, she has added too much detail into it and the excess should be eliminated, though she may feel she has less control over the project.

What will you say at the staff meeting?

The Work Breakdown Structure should be reviewed & all excess of information should be removed to prevent complexities in the WBS.

2. Distinguish between accuracy and precision, as it relates to estimation of project effort and cost.

If one is accurate, then the projected schedule end date is that near of the actual end date.

Precise? -2

Precise if close together

Accurate if close to real result

3. How can you determine if management in a software development organization is serious about risk management?

If a software company is serious about risk management, they should use a version control system like CVS or Subversion to keep track of previous versions of code. In the event the code needs to be reverted, it can be done. It also allows multiple people to work on the project at once,

−3

4. COCOMO is said to be a static single variable model. What does this mean?

Cocomo can be used on any type of software project to best calculate effort without needing to classify the project. There is only one variable that is needed, all other information remains constant.

what is it?

−1

$Effort = A\ Size^{B}$

5. In the early days of computing, software costs were a small part of the total system cost. Today, software costs are the largest component of total system cost. What are the implications of this shift for software engineering?

Computer hardware is now very cheap compared to 20-30 years ago, but software has remained the same because hardware is useless without software. Software costs have remained the same, but are a larger component than hardware now because of hardware's relative price decrease.

Not sure I believe this. The size of software written today >> size 50 years ago

6. Give 4 disadvantages of using lines-of-code as a metric of project size.

1. A poorly designed program may use an excess number of lines of code.
2. An estimate on the number of lines may not be what the actual number becomes.
3. If a class has many methods, not all of those methods might be used
4. A method may be used many times.

7. Steve McConnell recommends a top-ten risk list as **the** risk management tool. When the risk list is presented in table form, describe the contents of five columns.

Description of risk - short phrase about the risk

Risk discoverd - when the risk first appeared X

Risk duration - how long the risk has been on the list

Solution - how the risk should be fixed X

Owner - who should take care of the risk X

-3

This week priority

Last week

Duration

risk resolution status

8. Describe what is meant by an intractable problem in computer science.

-4

An unacceptable time to create a large data set

9. Give the 5 steps that are executed when performing risk analysis.

Define what the risk is

Determine its severity

prob of occurr

-3

Determine who or what the risk affects

Determine the cause of the risk

Determine who or what is responsible

3. Assign values to sev.

4. Select a priority technique.

5. Assign a priority to risk.

10. You have recently accepted a job with a software development company that has been in business for 6 years. List the five categories of estimation techniques and select the three that you think would be the most usable for your company. Give the advantages and disadvantages of each of the three.

These are all algorithmic

-4

1. Lines of code

advan. - can be easily computed

disadvan. - may not reflect flow of program, not accurate

2. Function points

advan. - reflects flow of program better than LOC

disadvan. - Some files, flows, or processes may have been looked over, skewing the calculations

3. Algorithmic Cost Modeling

advan. - can be adjusted to better suit the company or the project, easy to compute

disadvan. - not accurate

Algorithm, pricing to win, Parkinson, Expert review

Part 3 – Calculations, Analysis and Discussion

For the computations in this section, you must show all work, **or no credit will be given**. Point value for each question as indicated.

31/43

1. A software engineering project has 7 simple inputs, 3 average inputs, 2 complex outputs, 10 average inquiries, 4 complex master files, 2 transaction files, 2 simple interface, and 3 complex interfaces. The degree of influence factor for the project is 45. The weights are given in the table below. What is the number of function points? **(9 points)**

Category	Simple	Average	Complex
Inputs	3	5	9
Outputs	2	4	8
Inquiries	4	5	7
Master Files	7	10	20
Interfaces	6	8	10

? -1

$7(3) + 3(5) + 2(8) + 10(5) + 4(20) + 2(10) + 2(6) + 3(10)$

$\underbrace{21 + 15 + 16 + 50}_{102} + \underbrace{80 + 20 + 12 + 30}_{142}$

244

$= 0.65 + 0.01(244)$ TCF -1

FP -1

2. The FFP estimation method uses three components to estimate project size. List and describe the components and explain how they are used to estimate the project size. **(7 points)**

Files - files that the program interacts with

Flows - function calls & logic flow within the program -1

Processes - tasks that the program must complete

All of these are weighted equally. The sum of these three variables in the size.

flows - product interaction, GUI, I/O

3. You have estimated that an application will have 50 problem domain classes. The application will have a Complex Grapical User Interface (weight = 3). You know that your organization has a productivity constant of 16.5 (man-days/class). Estimate the effort in man-days required for the project. **(4 points)**

$X \text{ man-days} = 3(50 \text{ classes} \cdot 16.5 \frac{\text{man-days}}{\text{class}})$ $(50 + 3 * 50) * 16.5$

$2{,}475 \text{ man-days} = 3(825 \text{ man-days})$

-1

The questions on project scheduling refer to the tabular data and the accompanying Gantt chart.

ID	Task	Start	Duration (weeks)
1	Assess	1 Week after Kickoff Meeting	2
2	Design A	End of Assess	1
3	Implement A	End of Design A	3
4	Design B	End of Assess	2
5	Implement B	End of Design B	4
6	Design C	End of Design A, B	3
7	Implement C(subcontract)	End of Design C	6
8	Integrate A, B, C	Completion of A, B, C	2
9	System test	End of Integrate A, B, C	2

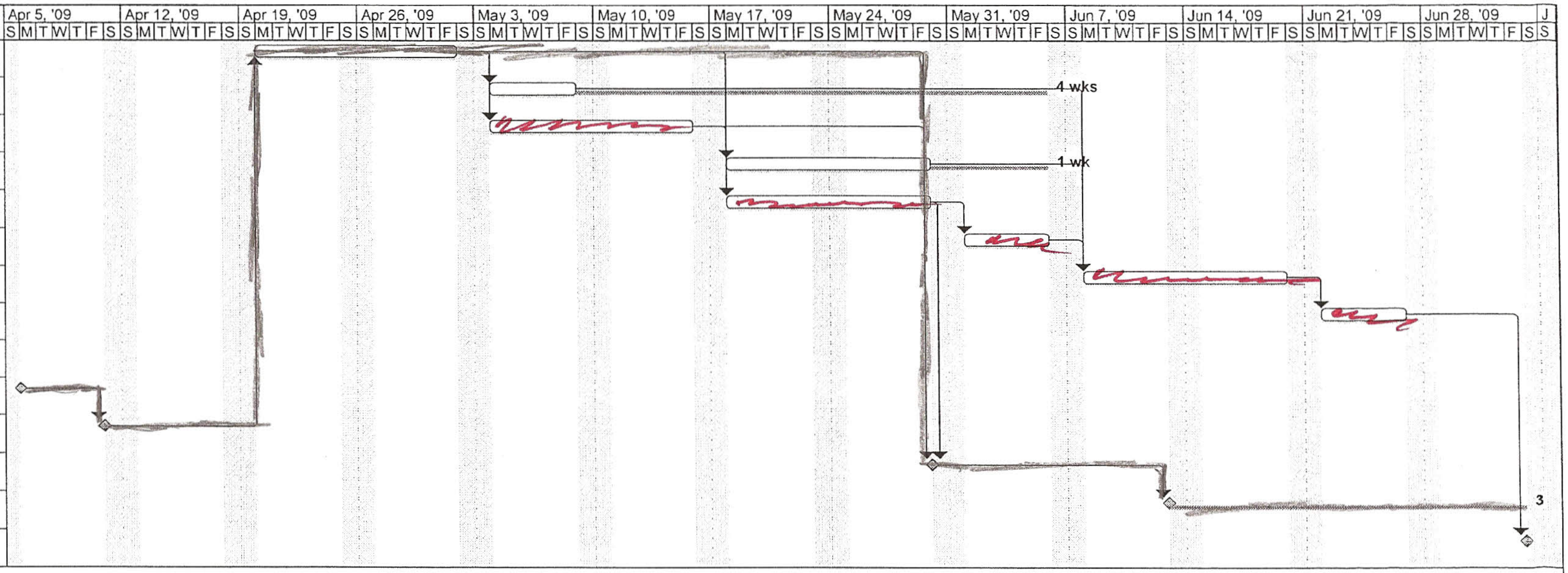
There are three events:

12. Project Award Date (= project start date): 7 April 2009 (Anticipated)
13. Project Kickoff Meeting with customer: 1 week after Project Award
14. Interim Design Review: When all designs are completed

There are two project deliverables:

15. Design Document: Deliver 2 weeks after Interim Design Review
16. System Delivery: 1 week after the end of system test

ID	Task Name
1	Design A
2	Implement A
3	Design B
4	Implement B
5	Design C
6	Implement C
7	Integerate A, B, C
8	System Test
9	
10	Project Awarded
11	Project Kickoff
12	Interim Design Reviewed
13	Design Doc Delivered
14	System Delivered



Project: Test Gantt.mpp
Date: Mon 3/23/09

Critical	Milestone	Rolled Up Critical	Progress
Critical Split	Slack	Rolled Up Critical Split	Milestone
Task	Slippage	Critical Split	
Split	Summary	Task	
Progress	Project Summary	Split	

4. Indicate the critical path on the Gantt chart by shading the appropriate task duration bars. **(5 points)**

-3

5. There are three errors on the Gantt chart. What are they? **(6 points)**

1. Task 9 has no name   there is NO task NINE actually shown
2. Task 11 - Project Kickoff should be first
3. Task 13 - Deliver Design Doc should come before the Designs of A, B, and C

The flow should always be diverted down or to the right.

6. Suppose that the Implement B task takes 1 additional week to complete. According to the Gantt chart, what is the effect on the system delivery date? **(3 points)**

none, there is ~~slippage~~ slack of 1 week

7. Consider the same schedule change in question 6. According to the Gantt chart, what is the effect on the critical path? **(3 points.)**

none

-3

8. For a particular project task, three different analysts have generated task duration times of 10 weeks, 3 weeks and 5 weeks. What task duration time would the conventional PERT technique provide? **(3 points)**

$$\frac{3 + 5(4) + 10}{6} = \frac{33}{6} = 5.5 \text{ weeks}$$

9. You give your supervisor the analyst's estimates and the PERT task duration times from Problem 8. Your supervisor "goes ballistic," saying, "I'll bet the long estimate was provided by Snerfle. That goldbricking loser always over-estimates a task." When you confirm that the high estimate was indeed Burf's, your boss suggests that you adjust the conventional PERT to give Burf's estimate one-half the weight that is assigned in the conventional PERT. Give the correct formula for the boss's suggestion. **(3 points)**

$$\text{task duration} = \frac{(\text{shortest time}) + 4(\text{most likely time}) + 0.5(\text{longest time})}{6 \quad 5.5}$$

-1